



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

April-2013

MA 203: Engineering Mathematics-IV

Date: 22.04.13, Monday Time: 01:30 p.m. To 04:30 p.m. Total: 70 Marks

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(4+3)

(a) Find Fourier Sine of the function $f(x) = \frac{\sin at}{t}$

(b) Find Fourier Transform of the function $f(x) = \begin{cases} 2, & \text{for } |x| < 2 \\ 0, & \text{for } |x| > 2 \end{cases}$

Q-2

(5+6+3)

- (a) Obtain an analytic function $f(z)$ in which $u - v = (x - y)(x^2 + 4xy + y^2)$
- (b) State and prove Cauchy-Riemann equations in polar form also find value of k such that the function $f(z) = r^2 \cos 2\theta + ir^2 \sin k\theta$ be an analytic function.
- (c) Under the transformation $w = iz + i$, show that the half plane $x > 0$ maps into the half plane $v > 1$ also sketch the image

OR

Q-2

(5+5+4)

- (a) Define Harmonic function, prove that the function $x^n - y^n$ is Harmonic if $n = 0, 1, 2$
- (b) Show that the function $f(z) = z^3 + iz$ is an analytic and find its derivative.
- (c) Define bilinear transformation, Find bilinear transformation which maps the points $(0, 1, \infty)$ onto $(-5, -1, 3)$

Q-3

(3+4+3+4)

(a) Find $L^{-1}\left[\frac{1}{s^2 + 3s + 2}\right]$ by method of Residues.

(b) Evaluate: $\int_0^{1+i} (x^2 + iy) dz$ along the path $y = x^2$

(c) Evaluate: $\int_{|z|=3} \frac{z^3 - z}{(z-2)^3} dz$

(d) Obtain Laurent's series expansion of function $\frac{1}{z-2}$ in the regions $|z| < 2$ and $|z| > 2$

OR

Q-3

(5+6+3)

(a) Find the Taylor series expansion for $\frac{1}{z^2 - 3z + 2}$ at point $z = 3$

(b) State Cauchy residue theorem and hence evaluate: $\int_{|z|=3} \frac{dz}{z^2(z+1)(z-1)}$

(c) State Cauchy integral formula and hence evaluate: $\int_{|z|=3} \frac{\cos z}{z} dz$

SECTION-II

Q-4

(3+4)

(a) Define unit step function and find its Z-transform

(b) Find Z-transform of $\sinh at$

Q-5

(4+5+5)

(a) Solve difference equation $f(t+1) - 2f(t) = 0$; $t \geq 1$, $f(0) = 1$

(b) Find $\log_{10} 656$ given that $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$, $\log_{10} 659 = 2.8189$ and $\log_{10} 661 = 2.8202$, using Langrange's interpolation formula.

(c) Find cubic polynomial which takes the following data:

x	0	1	2	3
$f(x)$	1	0	1	10

OR

Q-5

(7+7)

(a) Evaluate $\int_0^1 (1 + e^{2x} \sin 4x) dx$ using (i) Trapezoidal rule and (ii) Simpson's $\frac{1}{3^{rd}}$ rule

taking $h = \frac{1}{4}$

(b) Use Newton-Raphson method to solve the transcendental equation: $e^x = 5x$

Q-6

(7+7)

(a) Find by false position method the real root of the equation $\log_e x = \cos x$

(b) Apply the fourth order Runge-Kutta method to solve:

$\frac{dy}{dx} = xy$, for $x = 1.2$ initially, $y(1) = 2$. Take step size $h = 0.2$

OR

Q-6

(7+7)

(a) Using Picards method, solve the differential equation: $\frac{dy}{dx} = 1 + xy$, $y(0) = 1$ for

$x = 0.1$

(b) Find a real root of the equation $x^3 + x^2 - 1 = 0$ using bisection method.